

4. Vaccines still not commercially available (RTS, S/AS01 is the most advanced vaccine candidate against *Plasmodium falciparum*).

Helminths causing CNS diseases (I)

Objective ILOs

- List the most common helminthic **Cestodes** causing CNS diseases (Sparganosis, Cysticercosis, Hydatid disease, and Coenurosis).
- Recall the sources of infection, the infective and the diagnostic stages, of the selected parasitic diseases (Sparganosis, Cysticercosis, and Hydatid disease).
- Describe disease pathogenesis, pathological lesions and the main clinical presentations of each parasitic disease (Sparganosis, Cysticercosis, and Hydatid disease).
- Describe laboratory diagnosis, imaging and pathological studies of lesions caused by these parasites (Sparganosis, Cysticercosis, and Hydatid disease).
- Recall treatment and prevention of these parasites (Sparganosis, Cysticercosis, and Hydatid disease).

Helminthic <u>Cestodes</u> causing CNS diseases	▪ Sparganosis ▪ Cysticercosis ▪ Hydatid disease ▪ Coenurosis
Helminthic <u>Trematodes</u> causing CNS diseases	▪ <i>Heterophyes heterophyes</i> ▪ <i>Schistosoma</i> spp.
Helminthic <u>Nematodes</u> causing CNS diseases	▪ Visceral Larva Migrants (<i>Toxocara canis</i>, <i>Toxocara cati</i>) ▪ Heavy infection of <i>Ascaris lumbricoides</i>, <i>Ancylostoma duodenale</i>, <i>Strongyloides stercoralis</i>)

Neuro-cysticercosis

Definition	Invasion of human CNS tissues by <i>cysticercus cellulosae</i> , the larval stage of <i>Taenia solium</i> . Man acts as a blind intermediate host
Infective stage	Egg of <i>Taenia solium</i>
Diagnostic stage	<i>Cysticercus cellulosae</i>
Mode of infection	-<u>Heteroinfection</u>: Ingestion of food or water contaminated with eggs of <i>Taenia solium</i>. - <u>Autoinfection</u>: a. External autoinfection: Hand to mouth infection in a patient having adult worms in his intestine. b. Internal autoinfection: Regurgitation of detached gravid segments occurs by anti-peristaltic movement to the stomach, the released eggs descent again to the intestine where they hatch. The onchosphere from the egg penetrates intestinal mucosa to the circulation to various tissues where it develops into <i>cysticercus cellulosae</i>.

Pathology and clinical picture:

- The cyst vary in size from 0.5 to 2 cm in diameter and can be seen in meninges, ventricles or parenchyma.
- The number of cysts varies and range from a solitary cysticercus to several hundred.
- When present in the subarachnoid space and in the ventricle, the morphology of the cysticercus changes to form grape-like clusters (known as the “racemose” form of cysticercus).
- The clinical features vary depending on the location, size and number of the cysts and also the host’s immune response.
- **Asymptomatic:** most cases (80%).
- **Symptomatic:** (symptoms are mainly due to mass effects, inflammatory response or obstruction of foramina and ventricular system of the brain). seizures (most common manifestation), headache (unilateral or bilateral), focal neurological deficits, symptoms and signs of increase Intracranial pressure (nausea,

vomiting, altered mental status, visual changes, dizziness, cerebral edema), neurocognitive defects-learning disability, psychosis, depression).

Diagnosis:

1. History of diagnosis intestinal infection with *T. solium* (50% of cases).
2. The diagnosis is usually made on radiological imaging {CT (Computerized Tomography), MRI (Magnetic Resonance Image)}. (In case of solitary cysticercus granulomas, the lesion is seen as a single, small contrast-enhancing lesion).
3. Immunological tests:
 - Detection of anti-cysticercus antibodies in the CSF or serum by ELISA (in patients with reliable diagnosis of neuro-cysticercosis by imaging studies, serological test is not required since seropositivity depends on parasite load and endemicity, so could be false positive as in case of other parasitic infections or in patients from endemic area, or could be false negative in case of single lesion or calcification).
4. Molecular diagnosis: detect *T. solium* DNA in CSF
5. Biopsy: only in extreme cases.

Treatment: viability and location of the cyst should be confirmed.

1. Symptomatic (emergency care):

- Manage seizure activity.
- Monitor and correct metabolic abnormalities.
- Anti-convulsant drugs.
- In case of increased intracranial pressure (steroids, diuretics).

2. Specific anti-cysticercus agents: Albendazole

3. Surgery:

- If operable, surgical excision or endoscopic aspiration.
- For patients with hydrocephalus secondary to cysticercosis, ventricular shunt is done.

Prevention and control:

1. Treatment of patients infected with adult *T. solium*.
2. Protection of food or drink from contamination by human feces containing eggs.
3. Pure water supply and sanitary disposal of human excreta.
4. Personal hygiene to avoid auto-infection.
5. Avoid nauseating drugs to prevent the possibility of antiperistalsis (to avoid internal autoinfection).

Neuro-hydatosis

Definition	Invasion of human CNS tissue by larval stage (Hydatid cyst) of dog tapeworm <i>Echinococcus granulosus</i> (Man acts as a blind intermediate host)
Infective stage	Egg of <i>Echinococcus granulosus</i>
Diagnostic stage	Unilocular hydatid cyst
Mode of infection	Man becomes accidentally infected by: 1. Direct contact with or handling infected dogs, where the hairs are usually contaminated with eggs. 2. Ingestion of food or drink contaminated with eggs

Pathology and clinical picture:

- Cysts can reach up to 10cm in diameter.
- The wall is white and jelly-like, the appearance is likened to that of tender coconut flesh.
- The cyst could be unilocular or contain several daughter cysts and are surrounded by a fibrous wall formed by host tissues.
- Spilling of the cyst fluid due to trauma or surgery may trigger anaphylaxis as well as disseminated infection.

Clinical manifestations:

- Depending on the location and size of the cyst.
- Headache, seizures, focal neurological deficits, chorea, sign of spinal cord compression.

Diagnosis:

1. History of contact with dogs.
2. The diagnosis is usually made on radiological imaging (CT, MRI).
3. Immunological tests:
 - Microscopic examination of CSF: detection of hydatid sand.
 - Serological tests: detection of anti-hydatid antibodies by IHA, IFA, ELISA

Treatment:

- **Surgery:** if the cysts located in the operable sites
- **Medical treatment:** for in-operable and small sized operable cysts (Albendazole).

Prevention and control:

1. Regular treatment of infected pit dogs to reduce worm load.
2. Elimination of stray and infected dogs.
3. Prevention of dogs from eating infected offal's of domestic animals (cattle, sheep, ect.) in the slaughter houses.
4. Strict personal hygiene (e.g. such as washing hands after playing with or feeding the dogs).

Neuro-sparganosis

Definition	Invasion of human CNS tissue by <i>sparganum mansonii</i> , the larval stage of <i>Diphylllobothrium mansonii</i> .
Infective stage	Plerocercoid larva, Proceroid larva
Diagnostic stage	Plerocercoid larva
Mode of infection	1. Eating raw or undercooked flesh of frogs, snakes and birds infected with plerocercoid larva. 2. Applying infected flesh of frogs and snakes as foment or poultice to inflamed tissues as skin or eye. 3. Drinking water containing Cyclops infected with proceroid larva.

Pathology and clinical picture:

- The parasite burrows into the CNS and being long, forms multiple cavity lesions.
- The lesions are in the form of abscesses or cysts.
- The sparganum migrates by secreting proteases that degrade extra cellular matrix proteins.
- The parasite releases an allergic protease and prostaglandin E, which cleave immunoglobulin molecules in the host, thereby escaping detection.
- Dead and dying *sparganum* elicit an inflammatory reaction.
- Clinical manifestations: Depending on the location of the cyst. Focal neurological deficits, seizures and hemorrhage are seen depending on the location of the parasite.

Diagnosis:

1. History.

2. The diagnosis is usually made on radiological imaging (CT, MRI).
3. Immunological tests: Detection of anti-plerocercoids antibodies in the CSF or serum.
4. Rarely biopsy.

Treatment:

Surgical removal of the sparganum.

Prevention and control:

1. Proper cooking of flesh of the intermediate hosts (bird, snakes and frogs).
2. Avoidance of the use of frog tissue as poultices.
3. Destruction of Cyclops by boiling or filtering water before drinking.

Helminths causing CNS diseases (II)

Objective ILOs

- List the most common helminthic **Trematodes** as (*Heterophyes heterophyes*, *Schistosoma* sp.), and **Nematodes** as Visceral Larva Migrants (*Toxocara canis*, *Toxocara cati*, heavy infection of (*Ascaris lumbricoides*, *Ancylostoma duodenale*, *Strongyloides stercoralis*) causing CNS diseases.
- Demonstrate knowing the life cycle (definitive host, intermediate host, habitat) of the selected parasites (*Toxocara canis*, *Toxocara cati*).
- Identify the source of infection, the infective and diagnostic stages of the selected parasites (*Toxocara canis*, *Toxocara cati*).
- Describe disease pathogenesis and the main clinical presentations of selected parasites (*Toxocara canis*, *Toxocara cati*).
- Describe laboratory diagnosis, imaging and pathological studies of lesions caused by these parasites (*Toxocara canis*, *Toxocara cati*).
- Outline the general principle of treatment, prevention and control of these parasites (*Toxocara canis*, *Toxocara cati*).

Visceral Larva Migrants

1. Invasion of human viscera by larvae of dog and cat ascarids i.e. *Toxocara canis* and *Toxocara cati*.

Heavy infection of *Ascaris*, *Ancylostoma* or *Strongyloides*: Some larvae pass from the lung to left side of the heart, from where they pass to arterial circulation and settle in different organs producing Visceral Larva Migrants.

***Toxocara canis* (Dog ascarid), *Toxocara cati* (Cat ascarid)**

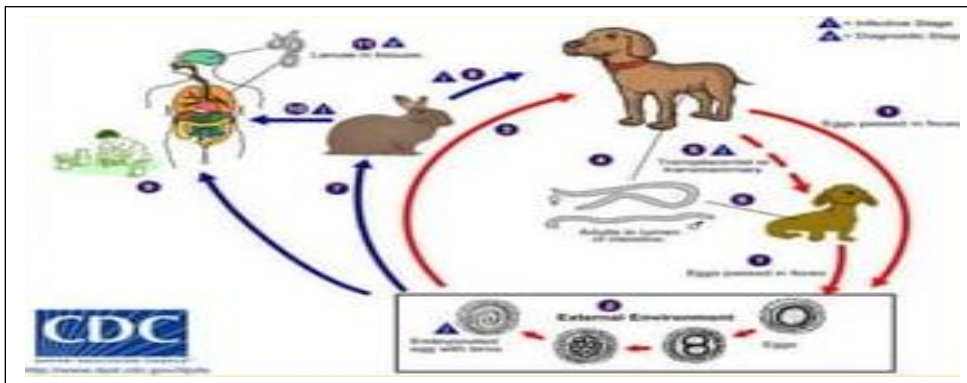
Mode of infection:

- Man is infected by ingestion of mature eggs (embryonated egg containing 2nd stage rhabditiform larva) with contaminated food, drinks or via hands contaminated with soil (most common in young children playing in contaminated soil or playground).

- Through the habit of eating earth (pica) reported by children.

Life cycle:

1. Adult worms live in the intestine of dogs and cats (not man).
2. Eggs pass in their feces.
3. Man is infected by ingestion of embryonated *Toxocara* eggs.
4. Larvae hatch, penetrate intestinal mucosa and are carried by blood stream to liver, lungs, brain, eyes and other organs where they wander causing inflammation with formation of eosinophilic granulomatous lesions.



Pathogenesis and clinical pictures:

The clinical picture depends on the site of such granuloma and the patient's allergic response may be:

1. **Asymptomatic** with persistent eosinophilia up to 80%.

2. Symptomatic infection:

- It is more common in children between 18 months and 4 years of age who has a history of close contact with the soil, dogs and cats.
- Persistent eosinophilia (20- 80 %).
- Fever, enlarged tender liver (larvae are arrested mostly in the liver).
- Pneumonitis and cough.
- Neuro-toxocariasis: Neurological manifestation as seizures.
- Ocular-toxocariasis: Typically occurs only in one eye and can cause irreversible vision loss. It is often presents as unilateral painless white retinal mass in a child, must be differentiated from retinoblastoma.
- Increased blood globulins specially gamma globulin.

Diagnosis:

I. Clinical diagnosis: Is more common in a child with chronic eosinophilia, has a history of exposure to dogs, cats or eating soil, hepatomegally or Hypergamma-globulinaemia is suggestive.

II. Laboratory diagnosis:

- Biopsy of accessible granuloma to detect the second stage rhabditiform larvae enclosed in granulomatous tissue.
- Eosinophilia 20- 80 %.
- Hypergamma-globulinaemia: Increased IgG, IgM and IgE.
- Serological tests: as IHA, IFA, ELISA (Antigen from the secretions of *Toxocara canis* larvae).
- Imaging techniques: Ultrasonography, CT, MRI.

Treatment:

1. Thiabendazole (Mintezol)
2. Corticosteroids in severe cases.

Prevention and control:

1. Avoid contamination of food, drink and hands with dog and cat feces.
2. Stray dogs and cats should be destroyed.
3. Pet dogs and cats should be de-wormed periodically.

Neuroschistosomiasis

Is one of the rare and most severe clinical presentations of the disease. It is caused by granuloma formation around eggs that lodge in the central nervous system, with *Schistosoma japonicum*, *Schistosoma mansoni* and very rare by *Schistosoma haematobium*.

Heterophyes heterophyes

The eggs may enter blood vessels or mesenteric lymph nodes and act as emboli to different tissues including brain, producing symptoms similar to cerebral hemorrhage or heart, causing myocarditis, chronic congestive heart failure, and death.

Parasitic infections of the eye

Objective ILOs